IDEATION PHASE — STRATEGIC FRAMEWORK

Problem-Solution Fit Canvas 2.0

Date: 29 October 2026 | **Team ID:** EV-BI-04 | **Project:** Electric Vehicle Market Performance Insights | **Max Marks:** 4 Marks

1. CUSTOMER SEGMENT(S) — CS

- Electric Vehicle Buyers
- Business Analysts
- Market Researchers
- Automobile Manufacturers
- Students and Researchers
- Government Policy Makers

2. JOBS-TO-BE-DONE / PROBLEMS — J&P

- Compare EV models side-by-side.
- Analyze EV historical sales trends.
- Compare battery technologies.
- Evaluate manufacturer performance.
- Study pricing & customer ratings.
- View sustainability insights on one platform.

3. TRIGGERS — TR

- Growing demand for Electric Vehicles.
- Need for business intelligence.
- Increase in EV market competition.
- Requirement for quick data-driven decisions.
- Interest in interactive dashboards.

4. EMOTIONS: BEFORE / AFTER — EM

😟 **BEFORE**

- Confused by scattered data
- Time-consuming analysis
- Difficult to compare datasets

🌟 **AFTER**

- Confident in decisions
- Easy decision making
- Interactive visualization

5. AVAILABLE SOLUTIONS — AS

EXISTING TOOLS:

- Excel & PDF Reports
- Government EV Reports
- Individual OEM Websites

LIMITATIONS:

- Static information
- Difficult comparison
- No interactive dashboard

6. CUSTOMER CONSTRAINTS — CC

- Lack of technical data knowledge.
- Limited access to centralized EV data.
- Time constraints.
- Difficulty understanding raw datasets.
- Navigating different information sources.

7. BEHAVIOUR — BE

- Search EV specs on Google.
- Compare manufacturer websites.
- Watch YouTube reviews.
- Read research reports.
- Explore Tableau dashboards.

8. CHANNELS OF BEHAVIOUR — CH

🌐 **ONLINE:**

- Flask App & Tableau Public
- Google Search, GitHub & YouTube

🏢 **OFFLINE:**

- College Projects & Papers
- EV Exhibitions & Meetings

9. PROBLEM ROOT CAUSE — RC

- **Data Fragmentation:** EV information is scattered across different platforms.
- **Lack of Centralized Visualization:** Absence of single unified dashboard.
- **Static Presentation:** Static reports reduce understanding and engagement.
- **Difficult Comparisons:** Hard to compare manufacturers & battery technologies.

10. YOUR SOLUTION (PROPOSED BI SOLUTION) — SL

Develop an Interactive EV Market Performance Insights Dashboard using Python Flask & Tableau Public.

- Market Performance Dashboard
- Business Intelligence Dashboard
- Research Dashboard
- Interactive Story Dashboard
- Responsive Flask Website
- Sustainability & CO_2 Analysis

PROJECT DESIGN PHASE-II — SYSTEM ENGINEERING

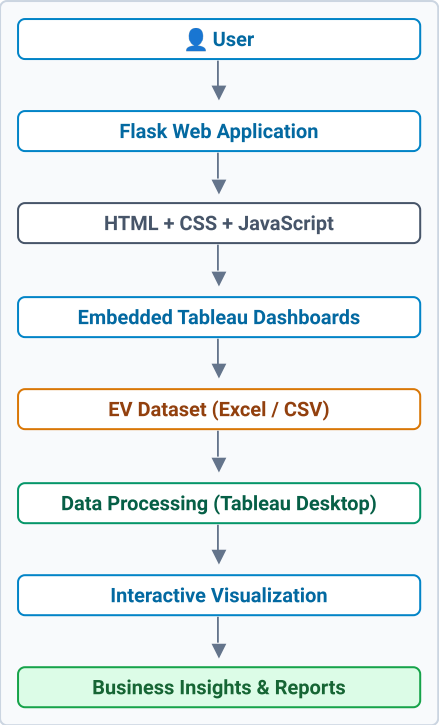
Solution Architecture & Component Breakdown

Project: EV Market Performance Insights Dashboard | **Team ID:** EV-BI-04

🧩 Component Description Table

Component	Description	Technology
User	Accesses the online web portal and explores dashboards.	Web Browser
Web Application	Hosts and serves the embedded dashboards.	Python Flask
Frontend UI	User interface providing responsive layout & navigation.	HTML, CSS, JavaScript
Dashboard	Provides interactive data visualizations & analytics.	Tableau Public
Dataset	Stores raw EV market data, specs, sales & ratings.	Excel / CSV
Data Processing	Handles data cleaning, transformation, and metrics calculation.	Tableau Desktop
Version Control	Manages application source code and revision history.	Git & GitHub
Output	Delivers final interactive business insights and visual stories.	Tableau Dashboard

📄 Architecture Flow



🌟 Expected Benefits

- ✔ Centralized EV analytics platform
- ✔ Easy comparison of EV models
- ✔ Improved data visualization
- ✔ Interactive business dashboards
- ✔ Better decision making support
- ✔ User-friendly web interface